

Bempedoic acid: effect of ATP-citrate lyase inhibition on low-density lipoprotein cholesterol and other lipids.

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Bempedoic acid is a new, first-in-class oral ATP-citrate lyase (ACLY) inhibitor that has to be converted to its CoA thioester before it inhibits ACLY. This conversion only occurs in the liver and not in skeletal muscle. This may explain why, unlike the statins, bempedoic acid does not cause myalgia. Bempedoic acid given at a dosage of 180 mg orally once daily produces a highly significant reduction in low-density lipoprotein cholesterol (LDL-C), non-high-density lipoprotein cholesterol, total cholesterol, apolipoprotein B and importantly also in high-sensitivity C-reactive protein. It has recently been approved by both the Food and Drug Administration (FDA) and the European Commission for use in adult patients with heterozygous familial hypercholesterolemia or atherosclerotic cardiovascular disease who require additional lowering of LDL-C, and for the treatment of adults with primary hypercholesterolemia (heterozygous familial and nonfamilial) or mixed dyslipidemia, respectively.

Bempedoic Acid, the First-in-Class Oral ATP Citrate Lyase Inhibitor with Hypocholesterolemic Activity: Clinical Pharmacology and Drug-Drug Interactions.

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Bempedoic acid is a new drug that improves the control of cholesterol levels, either as monotherapy or in combination with existing lipid-lowering therapies, and shows clinical efficacy in cardiovascular disease patients. Thus, patients with comorbidities and under multiple therapies may be eligible for bempedoic acid, thus facing the potential problem of drug-drug interactions (DDIs). Bempedoic acid is a prodrug administered orally at a fixed daily dose of 180 mg. The dicarboxylic acid is enzymatically activated by conjugation with coenzyme A (CoA) to form the pharmacologically active thioester (bempedoic acid-CoA). This process is catalyzed by very-long-chain acyl-CoA synthetase 1 (ACSVL1), expressed almost exclusively at the hepatic level. Bempedoic acid-CoA is a potent and selective inhibitor of ATP citrate lyase (ACL), a key enzyme in the biosynthetic pathway of cholesterol and fatty acids. The drug reduces low-density lipoprotein-cholesterol (LDL-C) (20-25%), non-high-density lipoprotein-cholesterol (HDL-C) (19%), apolipoprotein B (apoB) (15%), and total cholesterol (16%) in patients with hypercholesterolemia or mixed dyslipidemia. The drug has a favorable pharmacokinetics profile. Bempedoic acid and its metabolites are not substrates or inhibitors/inducers of cytochrome P450 (CYP450) involved in drug metabolism. On the other hand, bempedoic acid-glucuronide is a substrate for organic anion transporter 3 (OAT3). Bempedoic acid and its glucuronide are weak inhibitors of the OAT2, OAT3, and organic anion-transporting polypeptide 1B1 (OATP1B1) and 1B3 (OATP1B3). Thus, bempedoic acid could inhibit (perpetrator) the hepatic uptake of OATP1B1/3 substrate drugs and the renal elimination of OAT2 and OAT3 substrates and could suffer (victim) the effect of OAT3 transporter inhibitors, reducing its renal elimination. Based on these pharmacological characteristics, here, we describe the potential DDIs of bempedoic acid with concomitant medications and the possible clinical implications.

KHK, PNPLA3 and PPAR as Novel Targets for the Anti-Steatotic Action of Bempedoic Acid.

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Bempedoic acid (BemA) is an ATP-citrate lyase (ACLY) inhibitor used to treat hypercholesterolemia. We studied the anti-steatotic effect of BemA, and the mechanisms involved, in a model of fatty liver in female rats obtained through the administration of a high-fat diet supplemented with liquid fructose (HFHFr) for three months. In the third month, a group of rats was treated with BemA (30 mg/kg/day) by gavage. Plasma analytes, liver histology, adiposity, and the expression of key genes controlling fatty acid metabolism were determined, and PPAR agonism was explored by using luciferase reporter assays. Our results showed that, compared to HFHFr, BemA-treated rats exhibited lower body weight, higher liver/body weight, and reduced hepatic steatosis. In addition to ACLY inhibition, we found three novel mechanisms that could account for the anti-steatotic effect: (1) reduction of liver ketohexokinase, leading to lower fructose intake and reduced de novo lipogenesis; (2) increased expression of patatin-like phospholipase domain-containing protein 3, a protein related to the export of liver triglycerides to blood; and (3) PPAR α agonist activity, leading to increased hepatic fatty acid β -oxidation. In conclusion, BemA may represent a novel approach to treat hepatic steatosis, and therefore to avoid progression to advanced stages of non-alcoholic fatty liver disease.

Bempedoic acid and its role in contemporary management of hyperlipidemia in atherosclerosis.

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Atherosclerotic heart disease is the leading cause of mortality and morbidity in the USA. Low density lipoprotein (LDL) has been the target for many hypolipidemic agents to modify atherosclerotic risk. Bempedoic acid is a novel hypolipidemic drug that inhibits the enzymatic activity of ATP citrate lyase in the cholesterol synthesis pathway. CLEAR Harmony, CLEAR Wisdom, CLEAR Tranquillity and CLEAR Serenity have shown safety and efficacy associated with long term administration of this drug. Studies have shown effectiveness in reducing LDL-C in both statin intolerant patients and in patients on maximally tolerated doses of statin. The fixed drug combination of bempedoic acid and ezetimibe in a recent phase III showed significant reduction in LDL compared with placebo, which might be a promising future for LDL reduction among statin intolerant patients. Bempedoic acid also reduced inflammatory markers like hs-CRP. Given these results, bempedoic acid alone and in combination with ezetimibe received the USA FDA approval for adults with heterozygous familial hypercholesterolaemia or established atherosclerotic cardiovascular disease. We present a comprehensive review exploring the underlying mechanism, pre-clinical studies, and clinical trials of bempedoic acid and discuss the potential future role of the drug in treating hyperlipidaemia.

Bempedoic Acid: A New Non-statin Drug for the Treatment of Dyslipidemia.

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Statins are currently the first-line drugs for managing dyslipidemia due to their substantial clinical efficacy in reducing low-density lipoprotein cholesterol (LDL-C) and the risk of atherosclerotic cardiovascular disease (ASCVD). However, many patients do not reach their LDL-C target despite taking high-dose statins and some patients are intolerant of these drugs. Therefore, an additional or alternative pharmacological intervention may be required. Bempedoic acid is a novel lipid-lowering drug recently

approved for the treatment of dyslipidemia. This review describes the pharmacology of bempedoic acid and its clinical role in patients with dyslipidemia. Bempedoic acid, via its active coenzyme A (CoA) form, inhibits adenosine triphosphate (ATP)-citrate lyase, and reduces hepatic cholesterol synthesis through the mevalonate pathway. The reduction in plasma LDL-C by bempedoic acid is approximately 20%. In addition, this drug is able to lower the level of high-sensitivity C-reactive protein (hs-CRP) by 20%, which suggests anti-inflammatory activity. Bempedoic acid is well tolerated by the majority of patients. Possible common adverse drug reactions include upper respiratory tract infection, urinary tract infection and arthralgia. Serum creatinine and uric acid should be monitored since increased creatinine and hyperuricemia-associated new onset of gout and gout flares have been reported in patients taking bempedoic acid. Decreased hemoglobin levels and rare tendon ruptures have also been observed. Due to its efficacy and good safety profile, bempedoic acid might serve as a potential therapeutic alternative for the management of dyslipidemia.

Liver-specific ATP-citrate lyase inhibition by bempedoic acid decreases LDL-C and attenuates atherosclerosis.

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Despite widespread use of statins to reduce low-density lipoprotein cholesterol (LDL-C) and associated atherosclerotic cardiovascular risk, many patients do not achieve sufficient LDL-C lowering due to muscle-related side effects, indicating novel treatment strategies are required. Bempedoic acid (ETC-1002) is a small molecule intended to lower LDL-C in hypercholesterolemic patients, and has been previously shown to modulate both ATP-citrate lyase (ACL) and AMP-activated protein kinase (AMPK) activity in rodents. However, its mechanism for LDL-C lowering, efficacy in models of atherosclerosis and relevance in humans are unknown. Here we show that ETC-1002 is a prodrug that requires activation by very long-chain acyl-CoA synthetase-1 (ACSVL1) to modulate both targets, and that inhibition of ACL leads to LDL receptor upregulation, decreased LDL-C and attenuation of atherosclerosis, independently of AMPK. Furthermore, we demonstrate that the absence of ACSVL1 in skeletal muscle provides a mechanistic basis for ETC-1002 to potentially avoid the myotoxicity associated with statin therapy.

Exploring the Role of Bempedoic Acid in Metabolic Dysfunction Associated Steatotic Liver Disease: Actual Evidence and Future Perspectives.

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Metabolic dysfunction-associated steatotic liver disease (MASLD) involves excessive lipid accumulation in hepatocytes, impacting global healthcare due to its high prevalence and risk of progression to severe liver conditions. Its pathogenesis involves genetic, metabolic, and inflammatory factors, with cardiovascular events as the leading cause of mortality. This review examines the role of lipid-lowering therapies in MASLD, with a particular focus on bempedoic acid, a recently approved cholesterol-lowering agent for hypercholesterolemia and high cardiovascular-risk patients. It explores its potential in liver disease by modulating lipid metabolism and inflammatory pathways based on the most recent studies available. Bempedoic acid inhibits ATP-citrate lyase, reducing cholesterol and fatty acid synthesis while activating AMP-activated protein kinase to suppress gluconeogenesis and lipogenesis. Animal studies indicate its efficacy in reducing hepatic steatosis, inflammation, and fibrosis. Bempedoic acid holds

promise as a therapeutic for MASLD, offering dual benefits in lipid metabolism and inflammation. Further clinical trials are required to confirm its efficacy and safety in MASLD patients, potentially addressing the multifaceted nature of this disease.

Bempedoic acid in the management of lipid disorders and cardiovascular risk. 2023 position paper of the International Lipid Expert Panel (ILEP).

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Cardiovascular disease (CVD) is a chronic non-communicable disease (NCD) and the predominant cause of morbidity and mortality worldwide. Substantial reductions in the CVD prevalence have been achieved in recent years by the attenuation of risk factors (particularly hypertension and dyslipidaemias) in primary and secondary prevention. Despite the remarkable success of lipid lowering treatments, and of statins in particular, in reducing the risk of CVD, there is still an unmet clinical need for the attainment of guideline lipid-targets in even 2/3 of patients. Bempedoic acid, the first in-class inhibitor of ATP-citrate lyase presents a new approach to lipid-lowering therapy. By reducing the endogenous production of cholesterol, upstream of the rate-limiting enzyme HMG-CoA-reductase, i.e., the target of statins, bempedoic acid reduces circulating plasma concentrations of low-density lipoprotein cholesterol (LDL-C), and major adverse CVD events (MACE). Bempedoic acid has the potential to contribute to the reduction of CVD risk not only as monotherapy, but even further as part of a lipid-lowering combination therapy with ezetimibe, reducing LDL-C cholesterol up to 40%. This position paper of the International Lipid Expert Panel (ILEP) summarises the recent evidence around the efficacy and safety of bempedoic acid and presents practical recommendations for its use, which complement the 'lower-is-better-for-longer' approach to lipid management, which is applied across international guidelines for the management of CVD risk. Practical evidence-based guidance is provided relating to the use of bempedoic acid in atherosclerotic CVD, familial hypercholesterolaemia, and statin intolerance. Although there are still no sufficient data available for the role of bempedoic acid in the primary prevention of CVD, its favourable effects on plasma glucose and inflammatory markers makes this drug a rational choice in the patient-centred care of specific groups of primary prevention.

Adenosine Triphosphate Citrate Lyase and Fatty Acid Synthesis Inhibition: A Narrative Review.

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Adenosine triphosphate citrate lyase (ACLY) is a key regulatory enzyme of glucose metabolism, cholesterol and fatty acid synthesis, and the inflammatory cascade. Bempedoic acid, an ACLY inhibitor, significantly reduces atherogenic lipid markers, including low-density lipoprotein cholesterol (LDL-C), non-high-density lipoprotein cholesterol, and apolipoprotein B. Additional effects of ACLY inhibition include antitumor growth; reduction of triglycerides and proinflammatory molecules such as high-sensitivity C-reactive protein; less insulin resistance; reduction of hepatic lipogenesis; and weight loss. While numerous ACLY inhibitors have been identified, most of the clinical data have focused on bempedoic acid. The Cholesterol Lowering via Bempedoic Acid, an ACL-Inhibiting Regimen (CLEAR) program was a series of phase 3 clinical trials that evaluated its effects on lipid parameters and safety, leading to US Food and Drug Administration approval in 2020. CLEAR Outcomes was a phase 3,

double-blind, randomized, placebo-controlled trial in individuals with a history of statin intolerance, serum LDL-C level of 100 mg/dL or higher, and a history of, or at high risk for, cardiovascular disease. Bempedoic acid modestly reduced the primary 4-way cardiovascular composite end point as well as the individual components of myocardial infarction and coronary revascularization but did not reduce stroke, cardiovascular death, or all-cause mortality. Rates of gout and cholelithiasis were higher with bempedoic acid, and small increases in serum creatinine, uric acid, and hepatic-enzyme levels were also observed. ACLY inhibition with bempedoic acid has been established as a safe and effective therapy in high-risk patients who require further LDL-C lowering, particularly for those with a history of statin intolerance. The recently published CLEAR Outcomes trial revealed modest reductions in cardiovascular events with bempedoic acid, proportional to its LDL-C lowering, in high-risk individuals with statin intolerance and LDL-C levels of 100 mg/dL or higher. The additional effects of ACLY inhibition have prompted a more thorough search for novel ACLY inhibitors for conditions such as cancer, hypertriglyceridemia, chronic inflammation, type 2 diabetes, fatty liver disease, obesity, and metabolic syndrome. Similarly, therapies that reduce fatty acid synthesis are being explored for their use in cardiometabolic conditions.

Plasma proteomic associations with genetics and health in the UK Biobank.

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The Pharma Proteomics Project is a precompetitive biopharmaceutical consortium characterizing the plasma proteomic profiles of 54,219 UK Biobank participants. Here we provide a detailed summary of this initiative, including technical and biological validations, insights into proteomic disease signatures, and prediction modelling for various demographic and health indicators. We present comprehensive protein quantitative trait locus (pQTL) mapping of 2,923 proteins that identifies 14,287 primary genetic associations, of which 81% are previously undescribed, alongside ancestry-specific pQTL mapping in non-European individuals. The study provides an updated characterization of the genetic architecture of the plasma proteome, contextualized with projected pQTL discovery rates as sample sizes and proteomic assay coverages increase over time. We offer extensive insights into trans pQTLs across multiple biological domains, highlight genetic influences on ligand-receptor interactions and pathway perturbations across a diverse collection of cytokines and complement networks, and illustrate long-range epistatic effects of ABO blood group and FUT2 secretor status on proteins with gastrointestinal tissue-enriched expression. We demonstrate the utility of these data for drug discovery by extending the genetic proxied effects of protein targets, such as PCSK9, on additional endpoints, and disentangle specific genes and proteins perturbed at loci associated with COVID-19 susceptibility. This public-private partnership provides the scientific community with an open-access proteomics resource of considerable breadth and depth to help to elucidate the biological mechanisms underlying proteo-genomic discoveries and accelerate the development of biomarkers, predictive models and therapeutics¹.